Liyan Huang

liyanhua@usc.edu liyan-huang-b9776328a

Education

University of Southern California — Los Angeles, CA
Ph.D. in Computer Science — *Aug 2025 – present*

University of Southern California — Los Angeles, CA
M.S. in Computer Science — *Aug 2023 – May 2025*

Tsinghua University — Beijing, China
B.Eng. in Computer Science and Technology — *Aug 2019 – Jun 2023*

Research Experience

USC WebAssembly Group — *Aug 2023 - present*
Research Assistant, Advisor: ***Weihang Wang***

Research Interests: Software Engineering, Software Security.

- Applying program analysis and testing techniques to identify and mitigate security and reliability issues in software systems.
- Evaluating and enhancing the reliability of language models for code understanding and software engineering tasks.

Publications

CLARC: C/C++ Benchmark for Robust Code Search — *ICLR' 26*
Kaicheng Wang,* ***Liyan Huang****, Weike Fang, and Weihang Wang*

Broken Promise: Differential Analysis of Functional Discrepancies Between WebAssembly and Native Binaries — *WWW' 26*
Xiao Wu, Alan Romano*,* ***Liyan Huang****, Qiwen Yan, Cai Fu, and Weihang Wang*

Reusing Legacy Code in WebAssembly: Key Challenges of Cross-Compilation and Code Semantics Preservation — *SANER' 26*
Sara Baradaran, ***Liyan Huang****, Mukund Raghothaman, and Weihang Wang*

WaSCR: A WebAssembly Instruction-Timing Side Channel Repairer — *WWW' 25*
Liyan Huang*, Junzhou He*, Chao Wang, and Weihang Wang*

WBSan: WebAssembly Bug Detection for Sanitization and Binary-Only Fuzzing — *WWW' 25*
Xiao Wu, Junzhou He, ***Liyan Huang****, Cai Fu, and Weihang Wang*

(** Equal contribution.*)

Internship

ByteDance — Beijing, China
Software Engineer, Intern — *Sept 2022 - Feb 2023*

- Contributed to an internal platform backend, optimizing the usability of the system to enhance user experience and efficiency.

Academic Services

- Reviewer, ICML 2026.
- Program Committee Member (Security and Privacy, Research Track), WWW 2026.
- Reviewer, WWW 2025.
- Program Committee Member (Student Research Competition), ASE 2024.